

ADITHYA NATARAJAN

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SUMMARY

Software Engineer II at Intuit (4+ yrs) building AI-assisted security platforms and distributed backend services. Led Secure Design Agent (LangGraph) for automated threat modeling, built RAG knowledge services on OpenSearch, and delivered real-time abuse prevention systems (rate limiting, streaming, rules engine). Seeking Senior SWE roles in backend and AI platform infrastructure.

EDUCATION

M.Sc Computer Science	<i>Aug 2021 – Dec 2022</i>	B.Tech CS & Engineering	<i>July 2016 – May 2020</i>
Arizona State University, Tempe, AZ	GPA: 3.60	SASTRA University, India	GPA: 7.67/10

CERTIFICATIONS

AWS Certified Solutions Architect – Associate	<i>Jan 2026</i>
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TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript	AI/LLM: LangGraph, LangChain, RAG, Vector Search, OpenSearch, Langfuse
Backend: Spring Boot, FastAPI, Node.js/Express, Kafka, Drools	Data/Infra: AWS (Lambda, CDK, DynamoDB), PostgreSQL, Redis
Testing/CI: Cypress, WireMock, Jenkins, Docker, Kubernetes	

PROFESSIONAL EXPERIENCE

Intuit Inc	Mountain View, CA
Software Engineer 2	<i>Jan 2024 – Present</i>

- Architected and developed **Secure Design Agent**, a **multi-agent AI** system (LangGraph/Python) deployed across all product security reviews, orchestrating RAI Planner, Code Analysis, and Architecture Analysis agents with ReAct retry logic, reducing manual security review effort.
- Built **Reviewer Experience Service** (Java/Spring Boot) powering AI agent workflows: agent status tracking, prompt/version management, similarity scoring for threat assessments, and PostgreSQL-backed review lifecycle management.
- Developed **RAG Knowledge Service** (Python/FastAPI) with vector search, enabling semantic retrieval of security docs using embeddings and OpenSearch, improving consistency of AI threat assessments.
- Designed and shipped **Rate Shield**, a centralized rate limiting microservice (Spring Boot, DynamoDB, Bucket4j) protecting **5M+ daily** email transactions and SMS flows across QuickBooks surfaces.
- Architected and built **AppConnectGuard**, a real-time 3P abuse detection service processing **1-2M daily** events via Numaflow stream processing, integrating with STRA rules engine for automated app disconnect decisions.
- Led 3P abuse detection pipeline with NeLi (Negative List) auto-labeling and ThreatMetrix **ML-based** risk scoring integrated into STRA rules engine.
- Built **SpamShield** frontend plugin (TypeScript/React) for rate rule management; AWS Lambda Slack bot (CDK) for on-call reports via Splunk, improving incident visibility.

Software Engineer 1	<i>Jan 2023 – Jan 2024</i>
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- Architected end-to-end **SMS Screening** system (**100K+ daily**, **100 peak TPS**) with Bucket4j rate limiting, Drools eligibility workflows, and Kafka event publishing for QuickBooks risk decisions.
- Optimized QBDT latency via feature-flagged DynamoDB caching; pioneered **GenAI** integration for automated asset classification and review recommendations.
- Automated Jira workflows including priority assignment, watcher notifications, and lifecycle tracking, reducing manual triage overhead by **50%**.
- Built WireMock-based integration testing framework for downstream services; led JS to TypeScript migration and Cypress v12 upgrade, improving test reliability.

Software Engineer Intern	<i>May 2022 – Aug 2022</i>
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- Built security review portal (Java/Spring Boot, Thymeleaf) with AWS Athena integration for fraud analytics; added validation, tests, and admin-configurable mitigation rules to support review operations.

Freshworks Technologies Pvt Ltd	Chennai, India
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Software Engineer	<i>Oct 2020 – July 2021</i>
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- Developed Node.js backend for **Freshworks Marketplace** and shipped crash alerting system notifying developers when app crashes exceed thresholds; built REST APIs (App Installations, Custom Objects) using Express.js.
- Re-architected database schemas (MySQL, DynamoDB, Redis) reducing latency; published **Freshdesk SDK**, an NPM package enabling developers to integrate with Freshdesk APIs.

Software Engineer Intern	<i>Dec 2019 – Sep 2020</i>
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- Integrated SonarQube CI checks across repositories with unit tests (Node.js, Mocha.js); built DynamoDB table with TTL-based data retention for compliance.

PERSONAL PROJECTS

AI Short Form Content Agent (github.com/Adithya-nat/ai-reels-agent): Built AI video pipeline (FastAPI, MoviePy, FFmpeg) orchestrating GPT-4o scripts, DALL-E images, and TTS narration into short-form videos with WebSocket progress tracking.

Job Search Automation Tool (github.com/Adithya-nat/job-automation-tool): Full-stack app (React, Express, MongoDB) generating tailored resumes/cover letters via LLMs; integrated Puppeteer scraping and Slack job notifications.